CHAPTER 5

Continuous Random Variables

36 problems

17 textbook exercises · 11 strategic practice · 8 homework

Questions below; full solutions at the end of the chapter.

TEXTBOOK EXERCISES

Uniform and universality of the Uniform

Q1 Book Ch.5, Exercise 11 solution · p. 159

11. Ⓢ Let U be a Uniform r.v. on the interval $(-1, 1)$ (be careful about minus signs).

(a) Compute $E(U), \text{Var}(U)$, and $E(U^4)$.

(b) Find the CDF and PDF of U^2. Is the distribution of U^2 Uniform on $(0, 1)$?

Q2 Book Ch.5, Exercise 12 solution · p. 159

12. Ⓢ A stick is broken into two pieces, at a uniformly random break point. Find the CDF and average of the length of the longer piece.

Q3 Book Ch.5, Exercise 16 solution · p. 159

16. Ⓢ Let $U \sim \text{Unif}(0, 1)$, and

$$X = \log\left(\frac{U}{1-U}\right).$$

Then X has the Logistic distribution, as defined in Example 5.1.6.

(a) Write down (but do not compute) an integral giving $E(X^2)$.

(b) Find $E(X)$ without using calculus.

Hint: A useful symmetry property here is that $1 - U$ has the same distribution as U.

Q4 Book Ch.5, Exercise 19 solution · p. 159

19. Ⓢ Let F be a CDF which is continuous and strictly increasing. Let μ be the mean of the distribution. The quantile function, F^{-1}, has many applications in statistics and econometrics. Show that the area under the curve of the quantile function from 0 to 1 is μ.

    Hint: Use LOTUS and universality of the Uniform.

Normal

Q5 Book Ch.5, Exercise 32 solution · p. 160

32. Ⓢ Let $Z \sim \mathcal{N}(0,1)$ and let S be a random sign independent of Z, i.e., S is 1 with probability $1/2$ and -1 with probability $1/2$. Show that $SZ \sim \mathcal{N}(0,1)$.

Q6 Book Ch.5, Exercise 33 solution · p. 160

33. Ⓢ Let $Z \sim \mathcal{N}(0,1)$. Find $E(\Phi(Z))$ *without* using LOTUS, where Φ is the CDF of Z.

Q7 Book Ch.5, Exercise 34 solution · p. 160

34. Ⓢ Let $Z \sim \mathcal{N}(0,1)$ and $X = Z^2$. Then the distribution of X is called *Chi-Square with 1 degree of freedom*. This distribution appears in many statistical methods.

    (a) Find a good numerical approximation to $P(1 \le X \le 4)$ using facts about the Normal distribution, without querying a calculator/computer/table about values of the Normal CDF.

    (b) Let Φ and φ be the CDF and PDF of Z, respectively. Show that for any $t > 0$, $I(Z > t) \le (Z/t)I(Z > t)$. Using this and LOTUS, show that $\Phi(t) \ge 1 - \varphi(t)/t$.

Q8 Book Ch.5, Exercise 36 solution · p. 161

36. Ⓢ Let $Z \sim \mathcal{N}(0,1)$. A measuring device is used to observe Z, but the device can only handle positive values, and gives a reading of 0 if $Z \le 0$; this is an example of *censored data*. So assume that $X = ZI_{Z>0}$ is observed rather than Z, where $I_{Z>0}$ is the indicator of $Z > 0$. Find $E(X)$ and $\mathrm{Var}(X)$.

Exponential

Q9 Book Ch.5, Exercise 38 solution · p. 162

38. Ⓢ A post office has 2 clerks. Alice enters the post office while 2 other customers, Bob and Claire, are being served by the 2 clerks. She is next in line. Assume that the time a clerk spends serving a customer has the Exponential(λ) distribution.

    (a) What is the probability that Alice is the last of the 3 customers to be done being served?

    Hint: No integrals are needed.

    (b) What is the expected total time that Alice needs to spend at the post office?

Q10 Book Ch.5, Exercise 41

solution · p. 162

41. Ⓢ Fred wants to sell his car, after moving back to Blissville (where he is happy with the bus system). He decides to sell it to the first person to offer at least \$15,000 for it. Assume that the offers are independent Exponential random variables with mean \$10,000.

(a) Find the expected number of offers Fred will have.

(b) Find the expected amount of money that Fred will get for the car.

Q11 Book Ch.5, Exercise 44

solution · p. 162

44. Ⓢ Joe is waiting in continuous time for a book called *The Winds of Winter* to be released. Suppose that the waiting time T until news of the book's release is posted, measured in years relative to some starting point, has an Exponential distribution with $\lambda = 1/5$.

Joe is not so obsessive as to check multiple times a day; instead, he checks the website *once* at the end of each day. Therefore, he observes the day on which the news was posted, rather than the exact time T. Let X be this measurement, where $X = 0$ means that the news was posted within the first day (after the starting point), $X = 1$ means it was posted on the second day, etc. (assume that there are 365 days in a year). Find the PMF of X. Is this a named distribution that we have studied?

Q12 Book Ch.5, Exercise 50

solution · p. 163

50. Ⓢ Find $E(X^3)$ for $X \sim \text{Expo}(\lambda)$, using LOTUS and the fact that $E(X) = 1/\lambda$ and $\text{Var}(X) = 1/\lambda^2$, and integration by parts at most once. In the next chapter, we'll learn how to find $E(X^n)$ for all n.

Q13 Book Ch.5, Exercise 51

solution · p. 163

51. Ⓢ The *Gumbel distribution* is the distribution of $-\log X$ with $X \sim \text{Expo}(1)$.

(a) Find the CDF of the Gumbel distribution.

(b) Let $X_1, X_2, \dots$ be i.i.d. Expo(1) and let $M_n = \max(X_1, \dots, X_n)$. Show that $M_n - \log n$ converges in distribution to the Gumbel distribution, i.e., as $n \to \infty$ the CDF of $M_n - \log n$ converges to the Gumbel CDF.

Mixed practice

Q14 Book Ch.5, Exercise 55

solution · p. 163

55. Ⓢ Consider an experiment where we observe the value of a random variable X, and estimate the value of an unknown constant θ using some random variable $T = g(X)$ that is a function of X. The r.v. T is called an *estimator*. Think of X as the data observed in the experiment, and θ as an unknown parameter related to the distribution of X.

For example, consider the experiment of flipping a coin n times, where the coin has an unknown probability θ of Heads. After the experiment is performed, we have observed the value of $X \sim \text{Bin}(n, \theta)$. The most natural estimator for θ is then X/n.

The *bias* of an estimator T for θ is defined as $b(T) = E(T) - \theta$. The *mean squared error* is the average squared error when using $T(X)$ to estimate θ:

$$\text{MSE}(T) = E(T - \theta)^2.$$

Show that

$$\text{MSE}(T) = \text{Var}(T) + (b(T))^2 .$$

This implies that for fixed MSE, lower bias can only be attained at the cost of higher variance and vice versa; this is a form of the *bias-variance tradeoff*, a phenomenon which arises throughout statistics.

Q15 Book Ch.5, Exercise 56

solution · p. 163

56. Ⓢ (a) Suppose that we have a list of the populations of every country in the world.

Guess, without looking at data yet, what percentage of the populations have the digit 1 as their first digit (e.g., a country with a population of 1,234,567 has first digit 1 and a country with population 89,012,345 does not).

(b) After having done (a), look through a list of populations and count how many start with a 1. What percentage of countries is this? *Benford's law* states that in a very large variety of real-life data sets, the first digit approximately follows a particular distribution with about a 30% chance of a 1, an 18% chance of a 2, and in general

$$P(D = j) = \log_{10}\left(\frac{j+1}{j}\right), \text{ for } j \in \{1, 2, 3, \ldots, 9\},$$

where D is the first digit of a randomly chosen element. (Exercise from Chapter 3 asks for a proof that this is a valid PMF.) How closely does the percentage found in the data agree with that predicted by Benford's law?

(c) Suppose that we write the random value in some problem (e.g., the population of a random country) in scientific notation as $X \times 10^N$, where N is a nonnegative integer and $1 \leq X < 10$. Assume that X is a continuous r.v. with PDF

$$f(x) = c/x, \text{ for } 1 \leq x \leq 10,$$

and 0 otherwise, with c a constant. What is the value of c (be careful with the bases of logs)? Intuitively, we might hope that the distribution of X does not depend on the choice of units in which X is measured. To see whether this holds, let $Y = aX$ with $a > 0$. What is the PDF of Y (specifying where it is nonzero)?

(d) Show that if we have a random number $X \times 10^N$ (written in scientific notation) and X has the PDF f from (c), then the first digit (which is also the first digit of X) has Benford's law as PMF.

Hint: What does $D = j$ correspond to in terms of the values of X?

Q16 Book Ch.5, Exercise 57 solution · p. 164

57. Ⓢ (a) Let $X_1, X_2, \dots$ be independent $\mathcal{N}(0,4)$ r.v.s., and let J be the smallest value of j such that $X_j > 4$ (i.e., the index of the first X_j exceeding 4). In terms of Φ, find $E(J)$.

(b) Let f and g be PDFs with $f(x) > 0$ and $g(x) > 0$ for all x. Let X be a random variable with PDF f. Find the expected value of the ratio

$$R = \frac{g(X)}{f(X)}.$$

Such ratios come up very often in statistics, when working with a quantity known as a *likelihood ratio* and when using a computational technique known as *importance sampling*.

(c) Define

$$F(x) = e^{-e^{-x}}.$$

This is a CDF and is a continuous, strictly increasing function. Let X have CDF F, and define $W = F(X)$. What are the mean and variance of W?

Q17 Book Ch.5, Exercise 59 solution · p. 164

59. Ⓢ As in Example 5.7.3, athletes compete one at a time at the high jump. Let X_j be how high the jth jumper jumped, with $X_1, X_2, \dots$ i.i.d. with a continuous distribution. We say that the jth jumper is "best in recent memory" if he or she jumps higher than the previous 2 jumpers (for $j \geq 3$; the first 2 jumpers don't qualify).

(a) Find the expected number of best in recent memory jumpers among the 3rd through nth jumpers.

(b) Let A_j be the event that the jth jumper is the best in recent memory. Find $P(A_3 \cap A_4), P(A_3)$, and $P(A_4)$. Are A_3 and A_4 independent?

STRATEGIC PRACTICE

3 Continuous Distributions

Q18 SP 5 §3 · Problem 1 solution · p. 165

1. Let $Y = e^X$, where $X \sim \mathcal{N}(\mu, \sigma^2)$. Then Y is said to have a *LogNormal* distribution; this distribution is of great importance in economics, finance, and elsewhere. Find the CDF and PDF of Y (the CDF should be in terms of Φ).

Q19 SP 5 §3 · Problem 2 solution · p. 165

2. Let $U \sim \text{Unif}(0,1)$. Using U, construct a r.v. X whose PDF is $\lambda e^{-\lambda x}$ for $x > 0$ (and 0 otherwise), where $\lambda > 0$ is a constant. Then X is said to have a *Exponential* distribution; this distribution is of great importance in engineering, chemistry, survival analysis, and elsewhere.

Q20 SP 5 §3 · Problem 3

solution · p. 165

3. Let $Z \sim \mathcal{N}(0,1)$. Find $E(\Phi(Z))$ *without* using LOTUS, where Φ is the CDF of Z.

Q21 SP 5 §3 · Problem 4

solution · p. 166

4. A stick is broken into two pieces, at a uniformly random chosen break point. Find the CDF and average of the length of the longer piece.

4 LOTUS

Q22 SP 5 §4 · Problem 1

solution · p. 166

1. For $X \sim \text{Pois}(\lambda)$, find $E(X!)$ (the average factorial of X), if it is finite.

Q23 SP 5 §4 · Problem 2

solution · p. 166

2. Let $Z \sim \mathcal{N}(0,1)$. Find $E|Z|$.

Q24 SP 5 §4 · Problem 3

solution · p. 166

3. Let $X \sim \text{Geom}(p)$ and let t be a constant. Find $E(e^{tX})$, as a function of t (this is known as the *moment generating function*; we will see later how this function is useful).

1 Exponential Distribution and Memorylessness

Q25 SP 6 §1 · Problem 1

solution · p. 167

1. Fred (the protagonist of HW 6 #1) wants to sell his car, after moving back to Blissville (where he is happy with the bus system). He decides to sell it to the first person to offer at least \$15,000 for it. Assume that the offers are independent Exponential random variables with mean \$10,000.

   (a) Find the expected number of offers Fred will have.

   (b) Find the expected amount of money that Fred gets for the car.

Q26 SP 6 §1 · Problem 2

solution · p. 167

2. Find $E(X^3)$ for $X \sim \text{Expo}(\lambda)$, using LOTUS and the fact that $E(X) = 1/\lambda$ and $\text{Var}(X) = 1/\lambda^2$, and integration by parts at most once (see also Problem 1 in the MGF section).

Q27 SP 6 §1 · Problem 3

solution · p. 168

3. Let $X_1, \ldots, X_n$ be independent, with $X_j \sim \text{Expo}(\lambda_j)$. (They are i.i.d. if all the λ_j's are equal, but we are not assuming that.) Let $M = \min(X_1, \ldots, X_n)$. Show that $M \sim \text{Expo}(\lambda_1 + \cdots + \lambda_n)$, and interpret this intuitively.

Q28 SP 6 §1 · Problem 4

solution · p. 168

4. A post office has 2 clerks. Alice enters the post office while 2 other customers, Bob and Claire, are being served by the 2 clerks. She is next in line. Assume that the time a clerk spends serving a customer has the Exponential(λ) distribution.

   (a) What is the probability that Alice is the last of the 3 customers to be done being served Hint: no integrals are needed.

   (b) What is the expected total time that Alice needs to spend at the post office?

HOMEWORK

Q29 HW 5 · Problem 2

solution · p. 169

2. Joe is waiting in continuous time for a book called *The Winds of Winter* to be released. Suppose that the waiting time T until news of the book's release is posted, measured in years relative to some starting point, has PDF $\frac{1}{5}e^{-t/5}$ for $t > 0$ (and 0 otherwise); this is known as the *Exponential distribution* with parameter $1/5$. The news of the book's release will be posted on a certain website.

Joe is not so obsessive as to check multiple times a day; instead, he checks the website *once* at the end of each day. Therefore, he observes the day on which the news was posted, rather than the exact time T. Let X be this measurement, where $X = 0$ means that the news was posted within the first day (after the starting point), $X = 1$ means it was posted on the second day, etc. (assume that there are 365 days in a year). Find the PMF of X. Is this a distribution we have studied?

Q30 HW 5 · Problem 3

solution · p. 170

3. Let U be a Uniform r.v. on the interval $(-1, 1)$ (be careful about minus signs).

(a) Compute $E(U), \text{Var}(U)$, and $E(U^4)$.

(b) Find the CDF and PDF of U^2. Is the distribution of U^2 Uniform on $(0, 1)$?

Q31 HW 5 · Problem 4

solution · p. 170

4. Let F be a CDF which is continuous and strictly increasing. The inverse function, F^{-1}, is known as the *quantile function*, and has many applications in statistics and econometrics. Find the area under the curve of the quantile function from 0 to 1, in terms of the mean μ of the distribution F. Hint: Universality.

Q32 HW 5 · Problem 5 solution · p. 171

5. Let $Z \sim \mathcal{N}(0,1)$. A measuring device is used to observe Z, but the device can only handle positive values, and gives a reading of 0 if $Z \leq 0$; this is an example of *censored data*. So assume that $X = ZI_{Z>0}$ is observed rather than Z, where $I_{Z>0}$ is the indicator of $Z > 0$. Find $E(X)$ and $\text{Var}(X)$.

Q33 HW 6 · Problem 1 solution · p. 171

1. Fred lives in Blissville, where buses always arrive exactly on time, with the time between successive buses fixed at 10 minutes. Having lost his watch, he arrives at the bus stop at a random time (assume that buses run 24 hours a day, and that the time that Fred arrives is uniformly random on a particular day).

(a) What is the distribution of how long Fred has to wait for the next bus? What is the average time that Fred has to wait?

(b) Given that the bus has not yet arrived after 6 minutes, what is the probability that Fred will have to wait at least 3 more minutes?

(c) Fred moves to Blotchville, a city with inferior urban planning and where buses are much more erratic. Now, when any bus arrives, the time until the next bus arrives is an Exponential random variable with mean 10 minutes. Fred arrives at the bus stop at a random time, not knowing how long ago the previous bus came. What is the distribution of Fred's waiting time for the next bus? What is the average time that Fred has to wait? (Hint: don't forget the memoryless property.)

(d) When Fred complains to a friend how much worse transportation is in Blotchville, the friend says: "Stop whining so much! You arrive at a uniform instant between the previous bus arrival and the next bus arrival. The average length of that interval between buses is 10 minutes, but since you are equally likely to arrive at any time in that interval, your average waiting time is only 5 minutes."

Fred disagrees, both from experience and from solving Part (c) while waiting for the bus. Explain what (if anything) is wrong with the friend's reasoning.

Q34 HW 6 · Problem 2 solution · p. 173

2. Three Stat 110 students are working independently on this pset. All 3 start at 1 pm on a certain day, and each takes an Exponential time with mean 6 hours to complete this pset. What is the earliest time when all 3 students will have completed this pset, on average? (That is, *all* of the 3 students need to be done with this pset.)

Q35 HW 6 · Problem 4 solution · p. 173

4. (a) Suppose that we have a list of the populations of every country in the world.

Guess, without looking at data yet, what percentage of the populations have the digit 1 as their first digit (e.g., a country with a population of 1,234,567 has first digit 1 and a country with population 89,012,345 does not).

Note: (a) is a rare problem where the only way to lose points is to find out the right answer rather than guessing!

(b) After having done (a), look through a list of populations such as

`http://en.wikipedia.org/wiki/List_of_countries_by_population`

and count how many start with a 1. What percentage of countries is this?

(c) *Benford's Law* states that in a very large variety of real-life data sets, the first digit approximately follows a particular distribution with about a 30% chance of a 1, an 18% chance of a 2, and in general

$$P(D = j) = \log_{10}\left(\frac{j+1}{j}\right), \text{ for } j \in \{1, 2, 3, \ldots, 9\},$$

where D is the first digit of a randomly chosen element. Check that this is a PMF (using properties of logs, not with a calculator).

(d) Suppose that we write the random value in some problem (e.g., the population of a random country) in scientific notation as $X \times 10^N$, where N is a nonnegative integer and $1 \leq X < 10$. Assume that X is a continuous r.v. with PDF

$$f(x) = c/x, \text{ for } 1 \leq x \leq 10$$

(and 0 otherwise), with c a constant. What is the value of c (be careful with the bases of logs)? Intuitively, we might hope that the distribution of X does not depend on the choice of units in which X is measured. To see whether this holds, let $Y = aX$ with $a > 0$. What is the PDF of Y (specifying where it is nonzero)?

(e) Show that if we have a random number $X \times 10^N$ (written in scientific notation) and X has the PDF $f(x)$ from (d), then the first digit (which is also the first digit of X) has the PMF given in (c).

Hint: what does $D = j$ correspond to in terms of the values of X?

Q36 HW 7 · Problem 5 solution · p. 175

5. The bus company from Blissville decides to start service in Blotchville, sensing a promising business opportunity. Meanwhile, Fred has moved back to Blotchville, inspired by a close reading of *I Had Trouble in Getting to Solla Sollew*. Now when Fred arrives at the bus stop, either of two independent bus lines may come by (both of which take him home). The Blissville company's bus arrival times are exactly 10 minutes apart, whereas the time from one Blotchville company bus to the next is Expo($\frac{1}{10}$). Fred arrives at a uniformly random time on a certain day.

(a) What is the probability that the Blotchville company bus arrives first?

Hint: one good way is to use the continuous Law of Total Probability.

(b) What is the CDF of Fred's waiting time for a bus?

CHAPTER 5 · SOLUTIONS

S1 Book Ch.5, Exercise 11

question · p. 149

Solution:

(a) We have $E(U) = 0$ since the distribution is symmetric about 0. By LOTUS,

$$E(U^2) = \frac{1}{2}\int_{-1}^{1} u^2 du = \frac{1}{3}.$$

So $\text{Var}(U) = E(U^2) - (EU)^2 = E(U^2) = \frac{1}{3}$. Again by LOTUS,

$$E(U^4) = \frac{1}{2}\int_{-1}^{1} u^4 du = \frac{1}{5}.$$

(b) Let $G(t)$ be the CDF of U^2. Clearly $G(t) = 0$ for $t \leq 0$ and $G(t) = 1$ for $t \geq 1$, because $0 \leq U^2 \leq 1$. For $0 < t < 1$,

$$G(t) = P(U^2 \leq t) = P(-\sqrt{t} \leq U \leq \sqrt{t}) = \sqrt{t},$$

since the probability of U being in an interval in $(-1, 1)$ is proportional to its length. The PDF is $G'(t) = \frac{1}{2}t^{-1/2}$ for $0 < t < 1$ (and 0 otherwise). The distribution of U^2 is *not* Uniform on $(0, 1)$ as the PDF is not a constant on this interval (it is an example of a *Beta distribution*, an important distribution that is introduced in Chapter 8).

S2 Book Ch.5, Exercise 12

question · p. 149

Solution: We can assume the units are chosen so that the stick has length 1. Let L be the length of the longer piece, and let the break point be $U \sim \text{Unif}(0, 1)$. For any $l \in [1/2, 1]$, observe that $L < l$ is equivalent to $\{U < l \text{ and } 1 - U < l\}$, which can be written as $1 - l < U < l$. We can thus obtain L's CDF as

$$F_L(l) = P(L < l) = P(1 - l < U < l) = 2l - 1,$$

so $L \sim \text{Unif}(1/2, 1)$. In particular, $E(L) = 3/4$.

S3 Book Ch.5, Exercise 16

question · p. 149

Solution:

(a) By LOTUS,

$$E(X^2) = \int_0^1 \left(\log\left(\frac{u}{1-u}\right)\right)^2 du.$$

(b) By the symmetry property mentioned in the hint, $1 - U$ has the same distribution as U. So by linearity,

$$E(X) = E(\log U - \log(1 - U)) = E(\log U) - E(\log(1 - U)) = 0.$$

S4 Book Ch.5, Exercise 19

question · p. 150

Solution: We want to find $\int_0^1 F^{-1}(u)du$. Let $U \sim \text{Unif}(0,1)$ and $X = F^{-1}(U)$. By universality of the Uniform, $X \sim F$. By LOTUS,

$$\int_0^1 F^{-1}(u)du = E(F^{-1}(U)) = E(X) = \mu.$$

Equivalently, make the substitution $u = F(x)$, so $du = f(x)dx$, where f is the PDF of the distribution with CDF F. Then the integral becomes

$$\int_{-\infty}^{\infty} F^{-1}(F(x))f(x)dx = \int_{-\infty}^{\infty} xf(x)dx = \mu.$$

Sanity check: For the simple case that F is the Unif(0,1) CDF, which is $F(u) = u$ on $(0,1)$, we have $\int_0^1 F^{-1}(u)du = \int_0^1 udu = 1/2$, which is the mean of a Unif(0,1).

S5 Book Ch.5, Exercise 32

question · p. 150

Solution: Condition on S to find the CDF of SZ:

$$\begin{aligned}
P(SZ \le x) &= P(SZ \le x|S = 1)\frac{1}{2} + P(SZ \le x|S = -1)\frac{1}{2} \\
&= P(Z \le x)\frac{1}{2} + P(Z \ge -x)\frac{1}{2} \\
&= P(Z \le x)\frac{1}{2} + P(Z \le x)\frac{1}{2} \\
&= \Phi(x),
\end{aligned}$$

where the penultimate equality is by symmetry of the Normal.

S6 Book Ch.5, Exercise 33

question · p. 150

Solution: By universality of the Uniform, $F(X) \sim \text{Unif}(0,1)$ for any continuous random variable X with CDF F. Therefore, $E(\Phi(Z)) = 1/2$.

S7 Book Ch.5, Exercise 34

question · p. 150

Solution:

(a) By symmetry of the Normal,

$$P(1 \le Z^2 \le 4) = P(-2 \le Z \le -1 \text{ or } 1 \le Z \le 2) = 2P(1 \le Z \le 2) = 2(\Phi(2) - \Phi(1)).$$

By the 68-95-99.7% Rule, $P(-1 \le Z \le 1) \approx 0.68$. This says that 32% of the area under the Normal curve is outside of $[-1, 1]$, which by symmetry says that 16% is in $(1, \infty)$. So $\Phi(1) \approx 1 - 0.16 = 0.84$. Similarly, $P(-2 \le Z \le 2) \approx 0.95$ gives $\Phi(2) \approx 0.975$. In general, symmetry of the Normal implies that for any $t > 0$,

$$P(-t \le Z \le t) = \Phi(t) - \Phi(-t) = 2\Phi(t) - 1.$$

(b) The inequality $I(Z > t) \le (Z/t)I(Z > t)$ is true since if the indicator is 0 then both sides are 0, and if it is 1 then $Z/t > 1$. So

$$E(I(Z > t)) \le \frac{1}{t}E(ZI(Z > t)) = \frac{1}{t}\int_{-\infty}^{\infty} zI(z > t)\varphi(z)dz = \frac{1}{t}\int_{t}^{\infty} z\varphi(z)dz.$$

The integral can be done using a substitution: letting $u = z^2/2$, we have

$$\int ze^{-z^2/2}dz = \int e^{-u}du = -e^{-u} + C = -e^{-z^2/2} + C.$$

Thus,

$$P(Z > t) = E(I(Z > t)) \le \varphi(t)/t,$$

which proves the desired bound on $\Phi(t)$.

S8 Book Ch.5, Exercise 36 question · p. 150

Solution: By LOTUS,

$$E(X) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} I_{z>0}ze^{-z^2/2}dz = \frac{1}{\sqrt{2\pi}}\int_{0}^{\infty} ze^{-z^2/2}dz.$$

Letting $u = z^2/2$, we have

$$E(X) = \frac{1}{\sqrt{2\pi}}\int_{0}^{\infty} e^{-u}du = \frac{1}{\sqrt{2\pi}}.$$

To obtain the variance, note that

$$E(X^2) = \frac{1}{\sqrt{2\pi}}\int_{0}^{\infty} z^2e^{-z^2/2}dz = \frac{1}{2}\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} z^2e^{-z^2/2}dz = \frac{1}{2},$$

since a $\mathcal{N}(0, 1)$ r.v. has variance 1. Thus,

$$\text{Var}(X) = E(X^2) - (EX)^2 = \frac{1}{2} - \frac{1}{2\pi}.$$

Note that X is neither purely discrete nor purely continuous, since $X = 0$ with probability $1/2$ and $P(X = x) = 0$ for $x \ne 0$. So X has neither a PDF nor a PMF; but LOTUS still works, allowing us to work with the PDF of Z to study expected values of functions of Z.

Sanity check: The variance is positive, as it should be. It also makes sense that the variance is substantially less than 1 (which is the variance of Z), since we are reducing variability by making the r.v. 0 half the time, and making it nonnegative rather than roaming over the entire real line.

S9 Book Ch.5, Exercise 38

question · p. 150

Solution:

(a) Alice begins to be served when either Bob or Claire leaves. By the memoryless property, the additional time needed to serve whichever of Bob or Claire is still there is Expo(λ). The time it takes to serve Alice is also Expo(λ), so by symmetry the probability is $1/2$ that Alice is the last to be done being served.

(b) The expected time spent waiting in line is $\frac{1}{2\lambda}$, since the minimum of two independent Expo(λ) r.v.s is Expo(2λ) (by Example 5.6.3). The expected time spent being served is $\frac{1}{\lambda}$. So the expected total time is

$$\frac{1}{2\lambda} + \frac{1}{\lambda} = \frac{3}{2\lambda}.$$

S10 Book Ch.5, Exercise 41

question · p. 151

Solution:

(a) The offers on the car are i.i.d. $X_i \sim \text{Expo}(1/10^4)$. So the number of offers that are too low is Geom(p) with $p = P(X_i \geq 15000) = e^{-1.5}$. Including the successful offer, the expected number of offers is thus $(1-p)/p + 1 = 1/p = e^{1.5}$.

(b) Let N be the number of offers, so the sale price of the car is X_N. Note that

$$E(X_N) = E(X|X \geq 12000)$$

for $X \sim \text{Expo}(1/10^4)$, since the successful offer is an Exponential for which our information is that the value is at least \$15,000. To compute this, remember the memoryless

property! For any $a > 0$, if $X \sim \text{Expo}(\lambda)$ then the distribution of $X - a$ given $X > a$ is itself Expo(λ). So

$$E(X|X \geq 15000) = 15000 + E(X) = 25000,$$

which shows that Fred's expected sale price is \$25,000.

S11 Book Ch.5, Exercise 44

question · p. 151

Solution: The event $X = k$ is the same as the event $k \leq 365T < k+1$, i.e., $X = \lfloor 365T \rfloor$, where $\lfloor t \rfloor$ is the floor function of t (the greatest integer less than or equal to t). The CDF of T is $F_T(t) = 1 - e^{-t/5}$ for $t > 0$ (and 0 for $t \leq 0$). So

$$P(X = k) = P\left(\frac{k}{365} \leq T < \frac{k+1}{365}\right) = F_T\left(\frac{k+1}{365}\right) - F_T\left(\frac{k}{365}\right) = e^{-k/1825} - e^{-(k+1)/1825}.$$

This factors as $\left(e^{-1/1825}\right)^k (1 - e^{-1/1825})$, which shows that $X \sim \text{Geom}(1 - e^{-1/1825})$.

Sanity check: A Geometric distribution is plausible for a waiting time, and does take values $0, 1, 2, \ldots$. The parameter $p = 1 - e^{-1/1825} \approx 0.0005$ is very small, which reflects both the fact that there are a lot of days in a year (so each day is unlikely) and the fact that the author is not known for the celerity of his writing.

S12 Book Ch.5, Exercise 50

question · p. 151

Solution: By LOTUS,

$$E(X^3) = \int_0^\infty x^3 \lambda e^{-\lambda x} dx = -x^3 e^{-\lambda x}\Big|_0^\infty + \frac{3}{\lambda}\int_0^\infty x^2 \lambda e^{-\lambda x} dx$$
$$= \frac{3}{\lambda}E(X^2) = \frac{3}{\lambda}(Var(X) + (EX)^2) = \frac{6}{\lambda^3},$$

where the second equality uses integration by parts, letting $u = x^3$ and $dv = \lambda e^{-\lambda x} dx$ and we multiply the second term by 1 written as λ/λ.

S13 Book Ch.5, Exercise 51

question · p. 151

Solution:

(a) Let G be Gumbel and $X \sim \text{Expo}(1)$. The CDF is

$$P(G \le t) = P(-\log X \le t) = P(X \ge e^{-t}) = e^{-e^{-t}}$$

for all real t.

(b) The CDF of $M_n - \log n$ is

$$P(M_n - \log n \le t) = P(X_1 \le t + \log n, \dots, X_n \le t + \log n) = P(X_1 \le t + \log n)^n.$$

Using the Expo CDF and the fact that $(1 + \frac{x}{n})^n \to e^x$ as $n \to \infty$, this becomes

$$(1 - e^{-(t+\log n)})^n = (1 - \frac{e^{-t}}{n})^n \to e^{-e^{-t}}.$$

S14 Book Ch.5, Exercise 55

question · p. 151

Solution: Using the fact that adding a constant does not affect variance, we have

$$\begin{aligned}\text{Var}(T) &= \text{Var}(T - \theta)\\ &= E(T-\theta)^2 - (E(T-\theta))^2\\ &= \text{MSE}(T) - (b(T))^2,\end{aligned}$$

which proves the desired identity.

S15 Book Ch.5, Exercise 56

question · p. 152

Solution:

(a) What did you guess?

(b) According to Wikipedia (as of October 15, 2011), 63 out of 225 countries have a total population whose first digit is 1, which is 28%. (This depends slightly on whether certain territories included in the Wikipedia list should be considered as "countries" but the purpose of this problem is not to delve into the sovereignty or nation-status of territories). It is striking that 28% of the countries have first digit 1, as this is so much higher than one would expect from guessing that the first digit is equally likely to be any of $1, 2, \dots, 9$. This is an example of Benford's law; similar phenomena have been observed in many different settings (such as with lengths of rivers, physical constants, and stock prices).

(c) The PDF ($f(x) = c/x, 1 \leq x \leq 10$) must integrate to one, by definition; therefore

$$1 = c\int_1^{10} \frac{dx}{x} = c(\ln 10 - \ln 1) = c \ln 10.$$

So the constant of proportionality $c = 1/\ln 10 = \log_{10} e$). If $Y = aX$ (a *change in scale*), then Y has pdf c/y with the same value of c as before, except now $a \leq y \leq 10a$ rather than $1 \leq x \leq 10$. So the PDF takes the same form for aX as for X, but over a different range.

(d) The first digit $D = d$ when $d \leq X < d + 1$. The probability of this is

$$P(D = d) = P(d \leq X < d + 1) = \int_d^{d+1} \frac{1}{x \ln 10} dx,$$

which is $\log_{10}(d + 1) - \log_{10}(d)$, as desired.

S16 Book Ch.5, Exercise 57

question · p. 153

Solution:

(a) We have $J - 1 \sim \text{Geom}(p)$ with $p = P(X_1 > 4) = P(X_1/2 > 2) = 1 - \Phi(2)$, so $E(J) = 1/(1 - \Phi(2))$.

(b) By LOTUS,

$$E\frac{g(X)}{f(X)} = \int_{-\infty}^{\infty} \frac{g(x)}{f(x)} f(x)dx = \int_{-\infty}^{\infty} g(x)dx = 1.$$

(c) By universality of the Uniform, $W \sim \text{Unif}(0, 1)$. Alternatively, we can compute directly that the CDF of W is

$$P(W \leq w) = P(F(X) \leq w) = P(X \leq F^{-1}(w)) = F(F^{-1}(w)) = w$$

for $0 < w < 1$, so again we have $W \sim \text{Unif}(0, 1)$. Thus, $E(W) = 1/2$ and $\text{Var}(W) = 1/12$.

S17 Book Ch.5, Exercise 59

question · p. 153

Solution:

(a) Let I_j be the indicator of the jth jumper being best in recent memory, for each $j \geq 3$. By symmetry, $E(I_j) = 1/3$ (similarly to examples from class and the homework). By linearity, the desired expected value is $(n-2)/3$.

(b) The event $A_3 \cap A_4$ occurs if and only if the ranks of the first 4 jumps are $4, 3, 2, 1$ or $3, 4, 2, 1$ (where 1 denotes the best of the first 4 jumps, etc.). Since all orderings are equally likely,

$$P(A_3 \cap A_4) = \frac{2}{4!} = \frac{1}{12}.$$

As in (a), we have $P(A_3) = P(A_4) = 1/3$. So $P(A_3 \cap A_4) \neq P(A_3)P(A_4)$, which shows that A_3 and A_4 are not independent.

S18 SP 5 §3 · Problem 1

question · p. 153

1. Let $Y = e^X$, where $X \sim \mathcal{N}(\mu, \sigma^2)$. Then Y is said to have a *LogNormal* distribution; this distribution is of great importance in economics, finance, and elsewhere. Find the CDF and PDF of Y (the CDF should be in terms of Φ).

   The CDF of Y is

   $$P(Y \leq y) = P(X \leq \ln(y)) = P\left(\frac{X-\mu}{\sigma} \leq \frac{\ln(y)-\mu}{\sigma}\right) = \Phi\left(\frac{\ln(y)-\mu}{\sigma}\right),$$

   for $y > 0$ (and is 0 for $y \leq 0$). Taking the derivative using the chain rule, the PDF of Y is

   $$f(y) = \frac{1}{y\sigma\sqrt{2\pi}} e^{-(\ln(y)-\mu)^2/(2\sigma^2)}.$$

S19 SP 5 §3 · Problem 2

question · p. 153

2. Let $U \sim \text{Unif}(0, 1)$. Using U, construct a r.v. X whose PDF is $\lambda e^{-\lambda x}$ for $x > 0$ (and 0 otherwise), where $\lambda > 0$ is a constant. Then X is said to have a *Exponential* distribution; this distribution is of great importance in engineering, chemistry, survival analysis, and elsewhere.

   The CDF of X is given by $F(x) = \int_0^x \lambda e^{-\lambda t} dt = 1 - e^{-\lambda x}$, for $x > 0$ (and 0 otherwise). The inverse function is $F^{-1}(u) = -\lambda^{-1} \ln(1-u)$. So by Universality of the Uniform, $-\lambda^{-1} \ln(1-U)$ has CDF F.

S20 SP 5 §3 · Problem 3

question · p. 154

3. Let $Z \sim \mathcal{N}(0, 1)$. Find $E(\Phi(Z))$ *without* using LOTUS, where Φ is the CDF of Z.

   By Universality of the Uniform, $F(X) \sim \text{Unif}(0, 1)$ for any continuous random variable X with CDF F. Therefore, $E(\Phi(Z)) = 1/2$.

S21 SP 5 §3 · Problem 4 question · p. 154

4. A stick is broken into two pieces, at a uniformly random chosen break point. Find the CDF and average of the length of the longer piece.

   We can assume the units are chosen so that the stick has length 1. Let L be the length of the longer piece, and let the break point be $U \sim \text{Unif}(0,1)$. For any $l \in [1/2, 1]$, observe that $L < l$ is equivalent to $U < l, 1 - U < l$, which can be written as $1 - l < U < l$. We can thus obtain L's CDF as

$$F_L(l) \quad = \quad P(L < l) = P(1 - l < U < l) = 2l - 1,$$

   so $L \sim \text{Unif}(1/2, 1)$ and $E(L) = 3/4$.

S22 SP 5 §4 · Problem 1 question · p. 154

1. For $X \sim \text{Pois}(\lambda)$, find $E(X!)$ (the average factorial of X), if it is finite.

   By LOTUS,

$$E(X!) = e^{-\lambda} \sum_{k=0}^{\infty} k! \frac{\lambda^k}{k!} = \frac{e^{-\lambda}}{1 - \lambda},$$

   for $0 < \lambda < 1$ since this is a geometric series (and $E(X!)$ is infinite if $\lambda \geq 1$).

S23 SP 5 §4 · Problem 2 question · p. 154

2. Let $Z \sim \mathcal{N}(0,1)$. Find $E|Z|$.

   Let $f(z) = \Phi'(z)$ be the PDF of the $\mathcal{N}(0,1)$ distribution. By LOTUS and the substitution $u = z^2$ and since $|z|f(z)$ is an even function, we have

$$E|Z| \quad = \quad \int_{-\infty}^{\infty} |z| f(z) dz = 2 \int_0^{\infty} z f(z) dz = \int_0^{\infty} \frac{2ze^{-z^2/2}}{\sqrt{2\pi}} dz = \int_0^{\infty} \frac{e^{-u/2}}{\sqrt{2\pi}} du = \sqrt{2/}$$

S24 SP 5 §4 · Problem 3 question · p. 154

3. Let $X \sim \text{Geom}(p)$ and let t be a constant. Find $E(e^{tX})$, as a function of t (this is known as the *moment generating function*; we will see later how this function is useful).

   Letting $q = 1 - p$, we have

$$E(e^{tX}) = p \sum_{k=0}^{\infty} e^{tk} q^k = p \sum_{k=0}^{\infty} (qe^t)^k = \frac{p}{1 - qe^t},$$

   for $qe^t < 1$ (while for $qe^t \geq 1$, the series diverges).

1. Fred (the protagonist of HW 6 #1) wants to sell his car, after moving back to Blissville (where he is happy with the bus system). He decides to sell it to the first person to offer at least \$15,000 for it. Assume that the offers are independent Exponential random variables with mean \$10,000.

   (a) Find the expected number of offers Fred will have.

   The offers on the car are i.i.d. $X_i \sim \text{Expo}(1/10000)$. We are waiting for the first success, where "success" means that an offer is at least \$15,000. So the number of offers that are too low is $\text{Geom}(p)$ with $p = P(X_i \geq 15000) = \exp(-15000/10000) \approx 0.223$. Including the successful offer, the expected number of offers is thus $(1-p)/p + 1 = 1/p \approx 4.48$.

   (b) Find the expected amount of money that Fred gets for the car.

   Let N be the number of offers, so the sale price of the car is X_N. Note that

$$E(X_N) = E(X|X \geq 15000)$$

   for $X \sim \text{Expo}(1/10000)$, since the successful offer is an Exponential for which we have the information that the value is at least \$15,000. To compute this, remember the memoryless property of the Exponential! For any $a > 0$, if $X \sim \text{Expo}(\lambda)$ then the distribution of $X - a$ given $X > a$ is itself $\text{Expo}(\lambda)$. So

$$E(X|X \geq 15000) = 15000 + EX = \$25000,$$

   far more than his minimum price!

   Alternatively this can be done by integration (though we prefer the memoryless property), in which case it's crucial to use the conditional PDF (found using Bayes' Rule) rather than just integrating $xf(x)$ from a to ∞ (the conditional distribution needs to be properly normalized): letting $a = 15000$,

$$f(x|X>a) = \frac{P(X>a|X=x)f(x)}{P(X>a)} = \frac{\lambda e^{-\lambda x}}{e^{-\lambda a}} = \lambda e^{-\lambda(x-a)} \text{ for } x > a,$$

   and then

$$E(X|X \geq a) = \int_a^\infty x f(x|X>a)dx$$

   also works out to $\$25,000$.

2. Find $E(X^3)$ for $X \sim \text{Expo}(\lambda)$, using LOTUS and the fact that $E(X) = 1/\lambda$ and $\text{Var}(X) = 1/\lambda^2$, and integration by parts at most once (see also Problem 1 in the MGF section).

   By LOTUS, we have:

$$E(X^3) = \int_0^\infty x^3 \lambda e^{-\lambda x} dx = -x^3 e^{-\lambda x}\Big|_0^\infty + \frac{3}{\lambda}\int_0^\infty x^2 \lambda e^{-\lambda x} dx$$
$$= \frac{3}{\lambda}E(X^2) = \frac{3}{\lambda}(Var(X) + (EX)^2) = \frac{6}{\lambda^3},$$

   where the second equality uses integration by parts, letting $u = x^3$ and $dv = \lambda e^{-\lambda x}dx$ and we multiply the second term by 1 written as λ/λ.

S27 SP 6 §1 · Problem 3 question · p. 155

3. Let $X_1, \dots, X_n$ be independent, with $X_j \sim \text{Expo}(\lambda_j)$. (They are i.i.d. if all the λ_j's are equal, but we are not assuming that.) Let $M = \min(X_1, \dots, X_n)$. Show that $M \sim \text{Expo}(\lambda_1 + \dots + \lambda_n)$, and interpret this intuitively.

   We can find the distribution of M by considering its *survival function* $P(M > t)$, since the survival function is 1 minus the CDF.

$$P(M > t) = P(\min(X_1, \dots, X_n) > t) = P(X_1 > t, \dots, X_n > t)$$
$$= P(X_1 > t)\dots P(X_n > t) = e^{-\lambda_1 y} \cdot \dots \cdot e^{-\lambda_n y} = e^{-(\lambda_1 + \dots + \lambda_n)y},$$

   where the second equality holds since saying that the minimum of the X_j's is greater than t is the same as saying that all of the X_j's are greater than t. Thus, M has the survival function (and the CDF) of an Exponential distribution with parameter $\lambda_1 + \dots + \lambda_n$. Intuitively, it makes sense that M should have a continuous, memoryless distribution (which implies that it's Exponential) and if we interpret λ_j as rates, it makes sense that M has a combined rate of $\lambda_1 + \dots + \lambda_n$ since we can imagine, for example, X_1 as the waiting time for a green car to pass by, X_2 as the waiting time for a blue car to pass by, etc., assigning a color to each X_j; then M is the waiting time for a car with any of these colors to pass by.

S28 SP 6 §1 · Problem 4 question · p. 155

4. A post office has 2 clerks. Alice enters the post office while 2 other customers, Bob and Claire, are being served by the 2 clerks. She is next in line. Assume that the time a clerk spends serving a customer has the Exponential(λ) distribution.

(a) What is the probability that Alice is the last of the 3 customers to be done being served Hint: no integrals are needed.

Alice begins to be served when either Bob or Claire leaves. By the memoryless property, the additional time needed to serve whichever of Bob or Claire is still there is Exponential(λ). The time it takes to serve Alice is also Exponential(λ), so by symmetry the probability is 1/2 that Alice is the last to be done being served.

(b) What is the expected total time that Alice needs to spend at the post office?

The expected time spent waiting in line is $\frac{1}{2\lambda}$ since the minimum of two independent Exponentials is Exponential with rate parameter the sum of the two individual rate parameters. The expected time spent being served is $\frac{1}{\lambda}$. So the expected total time is

$$\frac{1}{2\lambda} + \frac{1}{\lambda} = \frac{3}{2\lambda}.$$

S29 HW 5 · Problem 2 question · p. 155

2. Joe is waiting in continuous time for a book called *The Winds of Winter* to be released. Suppose that the waiting time T until news of the book's release is posted, measured in years relative to some starting point, has PDF $\frac{1}{5}e^{-t/5}$ for $t > 0$ (and 0 otherwise); this is known as the *Exponential distribution* with parameter 1/5. The news of the book's release will be posted on a certain website.

Joe is not so obsessive as to check multiple times a day; instead, he checks the website *once* at the end of each day. Therefore, he observes the day on which the news was posted, rather than the exact time T. Let X be this measurement, where $X = 0$ means that the news was posted within the first day (after the starting point), $X = 1$ means it was posted on the second day, etc. (assume that there are 365 days in a year). Find the PMF of X. Is this a distribution we have studied?

The event $X = k$ is the same as the event $k \leq 365T < k + 1$, i.e., $X = \lfloor 365T \rfloor$, where $\lfloor t \rfloor$ is the floor function of t (the greatest integer less than or equal to t). The CDF of T is $F_T(t) = 1 - e^{-t/5}$ for $t > 0$ (and 0 for $t \leq 0$). So

$$P(X = k) = P\left(\frac{k}{365} \leq T < \frac{k+1}{365}\right) = F_T\left(\frac{k+1}{365}\right) - F_T\left(\frac{k}{365}\right) = e^{-k/1825} - e^{-(k+1)/1825}.$$

This factors as $\left(e^{-1/1825}\right)^k (1 - e^{-1/1825})$, which shows that $X \sim \text{Geom}(1 - e^{-1/1825})$.

Miracle check: A Geometric distribution is plausible for a waiting time, and does take values $0, 1, 2, \ldots$. The parameter $p = 1 - e^{-1/1825} \approx 0.0005$ is very small, which reflects both the fact that there are a lot of days in a year (so each day is unlikely) and the fact that the author is not known for the celerity of his writing.

S30 HW 5 · Problem 3 question · p. 155

3. Let U be a Uniform r.v. on the interval $(-1, 1)$ (be careful about minus signs).

(a) Compute $E(U), \mathrm{Var}(U)$, and $E(U^4)$.

We have $E(U) = 0$ since the distribution is symmetric about 0. By LOTUS,

$$E(U^2) = \frac{1}{2}\int_{-1}^{1} u^2 du = \frac{1}{3}.$$

So $\mathrm{Var}(U) = E(U^2) - (EU)^2 = E(U^2) = \frac{1}{3}$. Again by LOTUS,

$$E(U^4) = \frac{1}{2}\int_{-1}^{1} u^4 du = \frac{1}{5}.$$

(b) Find the CDF and PDF of U^2. Is the distribution of U^2 Uniform on $(0, 1)$?

Let $G(t)$ be the CDF of U^2. Clearly $G(t) = 0$ for $t \le 0$ and $G(t) = 1$ for $t \ge 1$, because $0 \le U^2 \le 1$. For $0 < t < 1$,

$$G(t) = P(U^2 \le t) = P(-\sqrt{t} \le U \le \sqrt{t}) = \sqrt{t},$$

since the probability of U being in an interval in $(-1, 1)$ is proportional to its length. The PDF is $G'(t) = \frac{1}{2}t^{-1/2}$ for $0 < t < 1$ (and 0 otherwise). The distribution of U^2 is *not* Uniform on $(0, 1)$ as the PDF is not a constant on this interval (it is an example of a *Beta distribution*, which is another important distribution in statistics and will be discussed later).

S31 HW 5 · Problem 4 question · p. 155

4. Let F be a CDF which is continuous and strictly increasing. The inverse function, F^{-1}, is known as the *quantile function*, and has many applications in statistics and econometrics. Find the area under the curve of the quantile function from 0 to 1, in terms of the mean μ of the distribution F. Hint: Universality.

We want to find $\int_0^1 F^{-1}(u)du$. Let $U \sim \mathrm{Unif}(0, 1)$ and $X = F^{-1}(U)$. By Universality of the Uniform, $X \sim F$. By LOTUS,

$$\int_0^1 F^{-1}(u)du = E(F^{-1}(U)) = E(X) = \mu.$$

Equivalently, make the substitution $u = F(x)$, so $du = f(x)dx$, where f is the PDF of the distribution with CDF F. Then the integral becomes

$$\int_{-\infty}^{\infty} F^{-1}(F(x))f(x)dx = \int_{-\infty}^{\infty} xf(x)dx = \mu.$$

Miracle check: For the simple case that F is the Unif$(0, 1)$ CDF, which is $F(u) = u$ on $(0, 1)$, we have $\int_0^1 F^{-1}(u)du = \int_0^1 udu = 1/2$, which is the mean of a Unif$(0, 1)$.

5. Let $Z \sim \mathcal{N}(0,1)$. A measuring device is used to observe Z, but the device can only handle positive values, and gives a reading of 0 if $Z \leq 0$; this is an example of *censored data*. So assume that $X = ZI_{Z>0}$ is observed rather than Z, where $I_{Z>0}$ is the indicator of $Z > 0$. Find $E(X)$ and $\text{Var}(X)$.

By LOTUS,

$$E(X) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} I_{z>0} z e^{-z^2/2} dz = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} z e^{-z^2/2} dz.$$

Letting $u = z^2/2$, we have

$$E(X) = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-u} du = \frac{1}{\sqrt{2\pi}}.$$

To obtain the variance, note that

$$E(X^2) = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} z^2 e^{-z^2/2} dz = \frac{1}{2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^2 e^{-z^2/2} dz = \frac{1}{2},$$

since a $\mathcal{N}(0,1)$ r.v. has variance 1. Thus,

$$\text{Var}(X) = E(X^2) - (EX)^2 = \frac{1}{2} - \frac{1}{2\pi}.$$

Note that X is neither purely discrete nor purely continuous, since $X = 0$ with probability $1/2$ and $P(X = x) = 0$ for $x \neq 0$. So X has neither a PDF nor a PMF; but LOTUS still works, allowing us to work with the PDF of Z to study expected values of functions of Z.

Miracle check: The variance is positive, as it should be. It also makes sense that the variance is substantially less than 1 (which is the variance of Z), since we are reducing variability by making the r.v. 0 half the time, and making it nonnegative rather than roaming over the entire real line.

S33 HW 6 · Problem 1

question · p. 156

1. Fred lives in Blissville, where buses always arrive exactly on time, with the time between successive buses fixed at 10 minutes. Having lost his watch, he arrives at the bus stop at a random time (assume that buses run 24 hours a day, and that the time that Fred arrives is uniformly random on a particular day).

(a) What is the distribution of how long Fred has to wait for the next bus? What is the average time that Fred has to wait?

The distribution is Uniform on $[0, 10]$, so the mean is 5 minutes.

(b) Given that the bus has not yet arrived after 6 minutes, what is the probability that Fred will have to wait at least 3 more minutes?

Let T be the waiting time. Then

$$P(T \geq 6 + 3 | T > 6) = \frac{P(T \geq 9, T > 6)}{P(T > 6)} = \frac{P(T \geq 9)}{P(T > 6)} = \frac{1/10}{4/10} = \frac{1}{4}.$$

(c) Fred moves to Blotchville, a city with inferior urban planning and where buses are much more erratic. Now, when any bus arrives, the time until the next bus arrives is an Exponential random variable with mean 10 minutes. Fred arrives at the bus stop at a random time, not knowing how long ago the previous bus came. What is the distribution of Fred's waiting time for the next bus? What is the average time that Fred has to wait? (Hint: don't forget the memoryless property.)

By the memoryless property, the distribution is Exponential with parameter $1/10$ (and mean 10 minutes) regardless of when Fred arrives (how much longer the next bus will take to arrive is independent of how long ago the previous bus arrived). The average time that Fred has to wait is 10 minutes.

(d) When Fred complains to a friend how much worse transportation is in Blotchville, the friend says: "Stop whining so much! You arrive at a uniform instant between the previous bus arrival and the next bus arrival. The average length of that interval between buses is 10 minutes, but since you are equally likely to arrive at any time in that interval, your average waiting time is only 5 minutes."

Fred disagrees, both from experience and from solving Part (c) while waiting for the bus. Explain what (if anything) is wrong with the friends reasoning.

The average length of a time interval between 2 buses is 10 minutes, but this does not imply that Fred's average waiting time is 5 minutes. This is because Fred is not equally likely to arrive at any of these intervals: Fred is more likely to arrive during a long interval between buses than to arrive during a short interval between buses. For example, if one interval between buses is 50 minutes and another interval is 5 minutes, then Fred is 10 times more likely to arrive during the 50 minute interval.

This phenomenon is known as *length-biasing*, and it comes up in many real-life situations. For example, asking randomly chosen mothers how many children they have yields a different distribution from asking randomly chosen people how many siblings they have, including themselves. Asking students the sizes of their classes and averaging those results may give a much higher value than taking a list of classes and averaging the sizes of each (this is called the *class size paradox*).

2. Three Stat 110 students are working independently on this pset. All 3 start at 1 pm on a certain day, and each takes an Exponential time with mean 6 hours to complete this pset. What is the earliest time when all 3 students will have completed this pset, on average? (That is, *all* of the 3 students need to be done with this pset.)

Label the students as $1, 2, 3$, and let X_j be how long it takes student j to finish the pset. Let T be the time it takes for all 3 students to complete the pset, so $T = T_1 + T_2 + T_3$ where $T_1 = \min(X_1, X_2, X_3)$ is how long it takes for one student to complete the pset, T_2 is the additional time it takes for a second student to complete the pset, and T_3 is the additional time until all 3 have completed the pset. Then $T_1 \sim \text{Expo}(\frac{3}{6})$ since, as shown on Strategic Practice 6, the minimum of independent Exponentials is Exponential with rate the sum of the rates. By the memoryless property, at the first time when a student completes the pset the other two students are "starting from fresh," so $T_2 \sim \text{Expo}(\frac{2}{6})$. Again by the memoryless property, $T_3 \sim \text{Expo}(\frac{1}{6})$. Thus, $E(T) = 2 + 3 + 6 = 11$, which shows that on average, the 3 students will have all completed the pset at midnight, 11 hours after they started.

4. (a) Suppose that we have a list of the populations of every country in the world.

Guess, without looking at data yet, what percentage of the populations have the digit 1 as their first digit (e.g., a country with a population of 1,234,567 has first digit 1 and a country with population 89,012,345 does not).

Note: (a) is a rare problem where the only way to lose points is to find out the right answer rather than guessing!

What did you guess for this part?

(b) After having done (a), look through a list of populations such as

`http://en.wikipedia.org/wiki/List_of_countries_by_population` and count how many start with a 1. What percentage of countries is this?

According to Wikipedia (as of October 15, 2011), 63 out of 225 countries have a total population whose first digit is 1, which is 28%. (This depends slightly on whether certain territories included in the Wikipedia list should be considered as "countries" but the purpose of this problem is not to delve into the sovereignty or nation-status of territories). It is striking that 28% of the countries have first digit 1, as this is so much higher than one would expect from guessing that the first digit is equally likely to be any of $1, 2, \ldots, 9$. This phenomenon is known as *Benford's Law* and a distribution similar to the one derived below has been observed in many diverse settings (such as lengths of rivers, physical constants, stock prices).

(c) *Benford's Law* states that in a very large variety of real-life data sets, the first digit approximately follows a particular distribution with about a 30% chance of a 1, an 18% chance of a 2, and in general

$$P(D = j) = \log_{10}\left(\frac{j+1}{j}\right), \text{ for } j \in \{1, 2, 3, \ldots, 9\},$$

where D is the first digit of a randomly chosen element. Check that this is a PMF (using properties of logs, not with a calculator).

The function $P(D = j)$ is nonnegative and the sum over all values is

$$\sum_{j=1}^{9} \log_{10} \frac{j+1}{j} = \sum_{j=1}^{9} (\log_{10}(j+1) - \log_{10}(j)).$$

All terms cancel except $\log_{10} 10 - \log_{10} 1 = 1$ (this is a *telescoping series*). Since the values add to 1 and are nonnegative, $P(D = j)$ is a PMF.

(d) Suppose that we write the random value in some problem (e.g., the population of a random country) in scientific notation as $X \times 10^N$, where N is a nonnegative integer and $1 \leq X < 10$. Assume that X is a continuous r.v. with PDF

$$f(x) = c/x, \text{ for } 1 \leq x \leq 10$$

(and 0 otherwise), with c a constant. What is the value of c (be careful with the bases of logs)? Intuitively, we might hope that the distribution of X does not depend on the choice of units in which X is measured. To see whether this holds, let $Y = aX$ with $a > 0$. What is the PDF of Y (specifying where it is nonzero)?

The PDF ($f(x) = c/x, 1 \leq x \leq 10$) must integrate to one, by definition; therefore

$$1 = c \int_{1}^{10} dx/x = c(\ln 10 - \ln 1) = c \ln 10.$$

So the constant of proportionality $c = 1/\ln 10 = \log_{10} e$). If $Y = aX$ (a *change in scale*), then Y has pdf c/y with the same value of c as before, except now $a \le y \le 10a$ rather than $1 \le x \le 10$. So the PDF takes the same form for aX as for X, but over a different range.

(e) Show that if we have a random number $X \times 10^N$ (written in scientific notation) and X has the PDF $f(x)$ from (d), then the first digit (which is also the first digit of X) has the PMF given in (c).

Hint: what does $D = j$ correspond to in terms of the values of X?

The first digit $D = d$ when $d \le X < d + 1$. The probability of this is

$$P(D = d) = P(d \le X < d + 1) = \int_d^{d+1} \frac{1}{x \ln 10} dx,$$

which is then $\log_{10}(d + 1) - \log_{10}(d)$, identical to our earlier PMF.

S36 HW 7 · Problem 5 question · p. 157

5. The bus company from Blissville decides to start service in Blotchville, sensing a promising business opportunity. Meanwhile, Fred has moved back to Blotchville, inspired by a close reading of *I Had Trouble in Getting to Solla Sollew*. Now when Fred arrives at the bus stop, either of two independent bus lines may come by (both of which take him home). The Blissville company's bus arrival times are exactly 10 minutes apart, whereas the time from one Blotchville company bus to the next is $\text{Expo}(\frac{1}{10})$. Fred arrives at a uniformly random time on a certain day.

(a) What is the probability that the Blotchville company bus arrives first?

Hint: one good way is to use the continuous Law of Total Probability.

Let $U \sim \text{Unif}(0, 10)$ be the arrival time of the next Blissville company bus, and $X \sim \text{Expo}(\frac{1}{10})$ be the arrival time of the next Blotchville company bus (the latter is $X \sim \text{Expo}(\frac{1}{10})$ by the memoryless property). Then

$$\begin{aligned} P(X < U) &= \int_0^{10} P(X < U|U = u)\frac{1}{10} du \\ &= \frac{1}{10}\int_0^{10} P(X < u|U = u) du \\ &= \frac{1}{10}\int_0^{10} (1 - e^{-u/10}) du = \frac{1}{e}. \end{aligned}$$

(b) What is the CDF of Fred's waiting time for a bus?

Let $T = \min(X, U)$ be the waiting time. Then

$$P(T > t) = P(X > t, U > t) = P(X > t)P(U > t).$$

So the CDF of T is

$$P(T \leq t) = 1 - P(X > t)P(U > t) = 1 - e^{-t/10}(1 - t/10),$$

for $0 < t < 10$ (and 0 for $t \leq 0$, and 1 for $t \geq 10$).